Local versus Global: Comparison of Convolutional and Transformer Models for Skin Lesion Classification with an Emphasis on Resources

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***Abstract*—**
***Index Terms*—**

I. Introduction

II. Relevant Literature

The potential of deep learning for skin cancer classification was demonstrated by Esteva et al. [1], who trained a convolutional neural network (CNN) on 129,450 clinical images covering 2,032 skin diseases. Their model achieved performance comparable to dermatologists on selected skin cancer classification tasks, highlighting the potential for accessible and cost-effective skin cancer detection.

The HAM10000 dataset, introduced by Tschandl et al. [2], contains 10,015 dermoscopic images across seven diagnostic categories. It provides a foundation for training and evaluating skin lesion classification models. However, its class imbalance presents a challenge when evaluating model performance.

A systematic review conducted by Kassem et al. [3] examined 53 machine learning and 49 deep learning methods for skin lesion classification. The authors identified major challenges such as limited datasets, image selection bias, and variations in skin color. These findings highlight the importance of reliable evaluation methods and appropriate data preprocessing.

Several studies have investigated the performance of CNNs and vision transformers (ViTs) for skin lesion classification. One such study [4] compared ResNet-152 and ViT on the HAM10000 dataset. The ViT achieved an accuracy of 98.28%, compared to 96.70% for ResNet-152. Although ViT achieved higher classification accuracy, the authors also highlighted its computational requirements and risk of overfitting.

A broader comparison of CNN and transformer-based models was conducted by Aruk et al. [5] using the HAM10000 and ISIC 2019 datasets. Their findings showed that transformer-based architectures can achieve competitive classification performance and that larger models do not necessarily outperform smaller ones. The study also investigated the effects of data augmentation and transfer learning.

The influence of lesion segmentation on classification performance was investigated by Mahbod et al. [6]. Their results showed that segmentation masks do not consistently improve classification performance. Removing the background could even reduce accuracy by discarding useful contextual information.

In addition to classification accuracy, computational efficiency is another important factor when comparing deep learning models. Ecer et al. [7] compared ResNet-50, EfficientNet-B3, and ViT-B/16 on the HAM10000 dataset. All three models achieved similar accuracy, ranging from 80.7% to 81.7%. However, ViT required substantially more energy, suggesting that higher computational costs do not necessarily lead to better classification performance.

Finally, a framework combining segmentation and classification was proposed by Fiaz et al. [8]. The authors used U-Net with a ResNet-34 encoder for skin lesion segmentation and EfficientNet-B0 for classification. On the HAM10000 dataset, their model achieved 85.57% classification accuracy and a Dice score of 0.8622 for segmentation. These results demonstrate the potential of combining segmentation and classification for skin lesion analysis.

References

[1] Esteva, A., Kuprel, B., Novoa, R. et al. Dermatologist-level classification of skin cancer with deep neural networks. *Nature* **542**, 115–118 (2017). https://doi.org/10.1038/nature21056

[2] Tschandl, P., Rosendahl, C. & Kittler, H. The HAM10000 dataset, a large collection of multi-source dermatoscopic images of common pigmented skin lesions. *Scientific Data* **5**, 180161 (2018). https://doi.org/10.1038/sdata.2018.161

[3] Kassem, M. A., Hosny, K. M., Damaševičius, R. & Eltoukhy, M. M. Machine learning and deep learning methods for skin lesion classification and diagnosis: A systematic review. *Diagnostics* **11**(8), 1390 (2021). https://doi.org/10.3390/diagnostics11081390

[4] Saputra, V. A., Devi, M. S. K., Diana & Kurniawan, A. Comparative analysis of convolutional neural networks and vision transformers for dermatological image classification. *Procedia Computer Science* **245**, 879–888 (2024). https://doi.org/10.1016/j.procs.2024.10.315

[5] Aruk, I., Pacal, I. & Toprak, A. N. A comprehensive comparison of convolutional neural network and visual transformer models on skin cancer classification. *Computational Biology and Chemistry* **120**, 108713 (2026). https://doi.org/10.1016/j.compbiolchem.2025.108713

[6] Mahbod, A., Tschandl, P., Langs, G., Ecker, R. & Ellinger, I. The effects of skin lesion segmentation on the performance of dermatoscopic image classification. *Computer Methods and Programs in Biomedicine* **197**, 105725 (2020). https://doi.org/10.1016/j.cmpb.2020.105725

[7] Ecer, N., Kankılıç, A. T., Karakoyun, Ö. et al. Diagnostic performance and energy consumption of deep learning models for dermoscopic image classification. *Discover Artificial Intelligence* **6**, 1530 (2026). https://doi.org/10.1007/s44163-026-02558-5

[8] Fiaz, M., Shoaib Khan, M. B., Khan, A. H., Bilal, A., Abdullah, M., Darem, A. A. & Sarwar, R. An explainable hybrid deep learning framework for precise skin lesion segmentation and multi-class classification. *Frontiers in Medicine* **12**, 1681542 (2025). https://doi.org/10.3389/fmed.2025.1681542